INCH-POUND

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DETAIL SPECIFICATION

SIMULATOR, LAUNCHING, ANTI-TANK, GUIDED MISSILE AND ROCKET M22 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts for and loading, assembling and packing for one type of simulator designated as Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22

Comments, suggestions, or questions on this document should be addressed to: U.S. Army CCDC AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1370

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3,4, and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3,4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Cards
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	DOD Standard Practice For Military Packaging
MIL-DTL-15011	-	Military Specification for Pallets, Material Handling, Wood Post Construction 4- Way Entry

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT CENTER DRAWINGS

11749562	–	Simulator, Antitank Missile Fire
12953563	–	Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22
9344296	–	Packing and Marking for Box, Packing Wood Wire Bound, for Box Fiberboard, for Simulator, Launching, AT, GM, and RKT, M22
9344297	–	Packing and Marking for Box, Fiberboard, for Simulator, Launching, AT, GM, and RKT, M22

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9344300	–	Box, Packing, Ammunition Wood, Wirebound
ACV00561	–	Unit Load Marking for Shipment and Storage, Ammunition and Explosives
ACV00831	–	Heat Treatment of Wooden Pallets, Skids, Filler Assemblies, and Dunnage Assemblies

(Copies of these drawings may be requested via e-mail at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil)

DEFESNE AMMUNITION CENTER

19-48-4116/107F	–	M22 Palletization
19-48-4116 Basic	–	Basic Palletization Drawing

(Copies of these drawings may be found at https://www.dau.edu/cop/ammo/Page/Default.aspx)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Materials. Materials and parts shall be in accordance with applicable drawings, specifications and standards.

3.3 Assembly. The simulator assembly shall comply with all requirements specified on Drawing 12953563, and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3.1 Contact closure switch. This contact closure switch shall be not greater than 1 ohm resistance when measured as specified in 4.5.4.

3.4 Moisture and volatile content.

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3.4.1 Signature and bang. The moisture and volatile content of these compositions at the time of loading shall be not greater than 1.00 percent.

3.5 Functioning. (Ambient condition, 21 ± 3°C or 70 ± 5°F, Hot condition, 51.5 ± 3°C or 125°F ± 5°F, Cold condition, -32 ± 3°C or -25 ±5°F). The assembly shall function and comply with the following requirements after conditioning at specified temperatures.

3.5.1 Sound characteristics. The sound level intensity of the cartridge functioning at a distance of 64 ± 1.0 feet (ft.), perpendicular to the direction of the output, shall be not less than 127 dB and not greater than 140 dB.

3.5.2 Functioning time. The time from application of direct current (VDC) to the firing device, with no current limitations, to audible report of 3.5.1 of the assembly shall be not greater than 0.110 seconds.

3.5.3 Signature characteristics. The cartridge shall produce a visible signature when fired from a launch tube. This signature shall be a minimum of six (6) feet in length, two (2) feet in width for a duration of 40 to 500 ms in winds of not more than 8 miles per hour. Residual smoke shall be white in appearance.

3.5.4 Crack case. There shall be no evidence of any cracks after functioning.

3.5.5 No-fire characteristics. If the M22 simulator does not function after two successfully firing signals, it shall be classified as a dud.

3.6 Vacuum chamber leak test. The cartridge shall show no evidence of air or water leakage when tested as specified in 4.5.6.

3.7 Jumble. The simulator shall comply with the following requirements when subjected to a jumble environment.

a. The simulator shall not function during test.
b. There shall be no loose pyrotechnic powder on the simulator or in the test box after jumbling.
c. The simulator components shall not separate during test.

3.8 Workmanship. All parts and assemblies shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

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3.9 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of verification		Classes of verification						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First article B – Conformance						
Section 3 & 5 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X	X	X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Materials	X		X	X	X	X	4.5.1
3.3	Assembly			X		X	X	4.3.3
3.3.1	Contact closure switch				X	X	X	4.5.4
3.4	Moisture and volatile content				X	X	X	4.5.2
3.4.1	Signature and bang				X	X	X	4.5.2.1
3.5	Functioning.				X	X	X	4.5.2
3.5.1	Sound characteristics				X	X	X	4.5.2.1
3.5.2	Functioning time				X	X	X	4.5.2.2
3.5.3	Signature characteristics				X	X	X	4.5.2.3
3.5.4	Crack Case				X	X	X	4.5.2.4
3.5.5	No-fire Characteristics				X	X	X	4.5.2.5
3.6	Leak test				X	X	X	4.4.5.1
3.7	Jumble				X	X		4.5.3
3.8	Workmanship			X		X	X	4.4.2.1 - 4.4.2.12
5.1.1	Packaging and marking for M22 ammunition box			X	X	X	X	4.3.3.10
5.1.2	Packing and marking wirebound box			X	X	X	X	4.3.3.11- 4.3.3.12
5.2.1	M22 Palletization			X	X	X		4.3.3.15
5.2.2	Palletized Load Marking			X	X	X	X	4.3.3.15
5.2.3	Wood Products Marking			X	X	X		4.3.3.14
5.3	Packaging workmanship			X		X	X	4.3.3.10 – 4.3.3.15

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4.1 Classification of inspections. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see paragraph 4.2)

b. Conformance Inspection (see paragraph 4.3)

4.2 First article inspection. When specified, a sample and its components and sub-assemblies shall be subjected to first article inspection in accordance to Table II.

4.2.1 First article quantity. The first article sample shall be from the first production lot produced by the contractor using the same production equipment, process procedures, and material supply sources as will be used in regular production. All successful leak tested units shall be used for functioning test and jumble test.

4.2.2 Inspections to be performed. The first article inspection shall be subjected to inspection and tests specified in Table II and 4.4.

4.2.3 Rejection. If any inspection fails to comply with any of the applicable requirements the first article quantity shall be rejected.

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TABLE II. First article acceptance inspection.

Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Case (Drawing 1174963) Examination for defects	5/mold cavity	3.3	4.4.2.1 &4.5.1
Tube (Drawing 11749635) Examination for defects	25	3.3	4.4.2.2
Closure Switch (Drawing 11749641) Examination for defects Resistance	10	3.3 3.3.1	4.4.2.3 4.5.4
Closure (Drawing 11749638) Examination for defects	5/mold cavity	3.3	4.4.2.4
Disc, Separator (Drawing 11749637)	25	3.3	4.5.1
Pellet Assembly (Drawing 11749633) Examination for defects	5	3.3	4.4.2.5
End Plate (Drawing 11749632) Examination for defects	5/mold cavity	3.3	4.4.2.6

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TABLE II. First article acceptance inspection continued.

Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Packing and marking (Drawing 9344297) Examination for defects Vacuum chamber leak test	 15 subpacks 	 5.1.1 3.6	 4.4.2.7 &4.5.1 4.4.2.10 - 4.4.2.12 4.5.7
Simulator, Launching, Guided Missile and Rocket, M22 (Drawing 12953563) Examination for defects	 175	 3.3	 4.4.2.8, 4.4.2.9
Functioning (Cold temperature) Sound characteristics Function time Flash Characteristics	50	3.5 3.5.1 3.5.2 3.3.3	4.5.2 4.5.2.1 4.5.2.2 4.5.2.3
Functioning (Hot temperature) Sound characteristics Function time Flash characteristics	50	3.5 3.5.1 3.5.2 3.5.3	4.5.2 4.5.2.1 4.5.2.2 4.5.2.3
Functioning (Ambient temperature) Sound characteristics Function time Flash characteristics	50	3.5 3.5.1 3.5.2 3.5.3	4.5.2 4.5.2.1 4.5.2.2 4.5.2.3
Jumble	25	3.7	4.5.3

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4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916. A lot shall consist of no more than one type of powder and no more than one granulation. In addition, each inspection lot of M22 assemblies shall contain:

a. Parts and compositions of one kind from one supplier.
b. Primers of one lot number.
c.
c. An ammunition lot number that is in accordance with MIL-STD-1168.

4.3.2 Classification of characteristics. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics shown in 3.2 through 3.8 when tested in accordance with 4.5 shall constitute minimum inspections to be performed.

4.3.3 Inspection requirements by classification of characteristics.

a. For the conformance inspection, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph “Acceptance by contractor-proposed provisions” of MIL-STD-1916).

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4.3.3.1	Title Case.		Dwg. No. 11749631	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	End plate seat depth to bottom groove	VL IV	3.3	Gage
102	End plate groove width	VL IV	3.3	Gage
103	Inside length	VL IV	3.3	Gage
104	Closure switch counterbore diameter	VL IV	3.3	Gage
105	Closure seat depth	VL IV	3.3	Gage
Minor				
201	Flange outer diameter	VL II	3.3	Gage
202	Small outer diameter	VL II	3.3	Gage
203	Poor workmanship	VL II	3.8	Visual

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4.3.3.2	Title Tube.		Dwg. No. 11749635	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Inside diameter, (min.)	VL IV	3.3	Gage
102	Outside diameter, (max.)	VL IV	3.3	Gage
Minor				
201	Total length	VL II	3.3	Gage
202	Poor workmanship	VL II	3.8	Visual

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4.3.3.3	Title Closure switch.		Dwg. No. 11749641	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Outside diameter, (max.)	VL IV	3.3	Gage
102	Resistance, (max.)	125, 0-1	3.3.1	4.5.4
Minor				
201	Poor workmanship	VL II	3.8	Visual

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4.3.3.4	Title Closure.		Dwg. No. 11749638	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Flange width	VL IV	3.3	Gage
102	Large diameter	VL IV	3.3	Gage
103	Small outside diameter	VL IV	3.3	Gage
104	Groove diameter	VL IV	3.3	Gage
Minor				
201	Poor workmanship	VL II	3.8	Visual

Notes.
See 4.5.1 for qualification and production control of mold cavities.

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4.3.3.5	Title Disc, separator.		Dwg. No. 11749637	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major 101	Diameter	VL IV	3.3	Gage
Minor 201	Poor workmanship	VL II	3.8	Visual

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4.3.3.6	Title Pellet assembly.		Dwg. No. 11749633	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major 101	Signature composition above flush	VL-IV	3.3	Gage
Minor 201	Signature composition greater than specified dimension below flush	VL II	3.3	Gage
202	Poor workmanship	VL II	3.8	Visual

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4.3.3.7	Title End plate.		Dwg. No. 11749632	Next Assy 12953563
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Inside diameter (primer hole)	VL IV	3.3	Gage
102	Large outside diameter	VL IV	3.3	Gage
103	Small outside diameter	VL IV	3.3	Gage
104	Depth of primer hole	VL IV	3.3	Gage
105	Overall thickness	VL IV	3.3	Gage
Minor				
201	Poor workmanship	VL II	3.8	Visual
Notes. See 4.5.1 for qualification and production control of mold cavities				

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4.3.3.8	Title Simulator, launching, AT, GM and RKT M22 (prior to assy of Closure Charge Cover).		Dwg. No. 12953563	Next Assy ---
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Moisture content of bang	1/	3.4.1	4.5.2.1
102	composition Weight of bang composition (verification)	2/	3.3	Balance
Minor				
201	Poor workmanship	VL II	3.8	Visual

Notes.

1. See para. 4.4.3.1
2. A sample of fifty (50) loading assemblies, prior to sealing the closure shall be selected from each eight (8) hours of production and subjected to a weight of charge verification as follows: Using a tared bang composition container, the weight difference due to the addition of the bang composition shall be determined. If the weight of any charge fails to meet the applicable drawing requirement, those assemblies loaded since the last acceptance verification test shall be classified defective and removed from the lot (Non-destructive test).

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4.3.3.9	Title Simulator, launching, AT, GM and RKT M22.	Dwg. No. 12953563	Next Assy 9344297

Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical				
1	Evidence of loose powder on exterior of assembly	100% & VL VII	3.3	Visual
Major				
101	Assembly damaged to extent that powder may leak	VL IV	3.3	Visual
102	Primer above flush or greater than max below flush	VL IV	3.3	Gage
103	Closure switch above flush or loose	VL IV	3.3	Gage
104	Any component (s) loose at any joint or contact	VL IV	3.3	Visual/Manual
105	Epoxy sealant missing, peeling or incorrectly applied	VL IV	3.3	Visual
106	Closure switch conductive area separated or torn	VL IV	3.3	Visual
107	Total length	VL IV	3.3	Gage

Notes.

(a) After three (3) consecutive lots have been accepted, sample may be reduced in accordance with para. 4.4.5.

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4.3.3.9	Title Simulator, launching, AT, GM and RKT M22 (continued).	.	Dwg. No 12953563	Next Assy 9344297

Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Major				
108	Leak Test	150 (a), 2-3	3.6	4.5.6
109	Functioning (Hot)	50 (a), 2-3	3.5	4.4.4
	Sound characteristics		3.5.1	4.5.2.1
	Functioning time		3.5.2	4.5.2.2
	Signature characteristics		3.5.3	4.5.2.3
110	Functioning (Cold)	50 (a), 3-4	3.5	4.4.4
	Sound characteristics		3.5.1	4.5.2.1
	Functioning time		3.5.2	4.5.2.2
	Signature characteristics		3.5.3	4.5.2.3
111	Functioning (Ambient)	50 (a), 2-3	3.5	4.4.4
	Sound characteristics		3.5.1	4.5.2.1
	Functioning time		3.5.2	4.5.2.2
	Signature characteristics		3.5.3	4.5.2.3
112	Cracked case	100%	3.5.4	Visual
113	No-fire characteristics		3.5.5	4.5.2.5
Minor				
201	Marking, missing, incorrect or illegible	VL II	3.3	Visual
202	Poor workmanship	VL II	3.8	Visual

Notes.
(a) After three (3) consecutive lots have been accepted, sample may be reduced in accordance with para. 4.4.5

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4.3.3.10	Title Packing and Marking for Box, Fiberboard, Simulator, launching, AT, GM and RKT, M22, prior to sealing.		Dwg. No. 9344297	Next Assy 9344296
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Verify primer end up and quantity	VL IV	5.1.1	Visual
102	Instruction sheet missing	VL IV	5.1.1	Visual
103	Paper moisture content	4.4.3.1	5.1.1	4.5.2.2
Minor				
201	Loose fit of partition to box and orientation of cutouts incorrect	VL II	5.1.1	Manual
202	Loose pack	VL II	5.1.1	Visual
203	Filler sheet missing	VL II	5.1.1	Visual
204	Marking missing, illegible, or incorrect	VL II	5.1.1	Visual
205	Poor workmanship	VL II	5.3	Visual

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4.3.3.11	Title Packing and Marking For Simulator, Launching, At, Gm and RKT, M22 in Box, Fiberboard (closure of barrier bag)		Dwg. No 9344297	Next Assy 9344296
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Barrier bag ruptured or heat seal incorrect	10, 0-1	5.1.1	4.5.8
102	Heat seal test	10, 0-1	5.1.1	4.5.7
Minor				
201	Barrier bag fold incorrect	VL II	5.1.1	Visual
202	Heat seal of barrier bag located so that one or more subsequent seals can be applied	VL II	5.1.1	Visual
203	Marking incorrect or not legible	VL II	5.1.1	Visual
204	Poor workmanship	VL II	5.3	Visual
Notes.				

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4.3.3.12	Title Packaging and Marking for Box, Fiberboard (after sealing)		Dwg. No. 9344297	Next Assy 9344296
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly	1/	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location		5.1.2	Visual
Minor				
201	Contents loose		5.1	Visual
202	Workmanship		5.3	Visual
Notes. 1/ Sample size and acceptance criteria shall be in accordance with Table III.				

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4.3.3.13	Title Packing and Marking for Box, Packing, Wood, For Box, Fiberboard For Simulator, launching, AT, GM and RKT, M22.		Dwg. No. 9344296	Next Assy ---
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Seal missing	VL IV	5.1.2	Visual
102	Incorrect number of fiberboard boxes	VL IV	5.1.2	Visual/Manual
Minor				
201	Bottom, sides or end fillers missing	VL II	5.1.2	Visual
202	Top insert filler missing	VL II	5.1.2	Visual
203	Marking incorrect or not legible	VL II	5.1.2	Visual
204	Loose pack	VL II	5.1.2	Visual/Manual
205	Poor workmanship	VL II	5.3	Visual

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4.3.3.14	Title Packaging and Marking for Box, WB (after sealing)		Dwg. No. 9344297	Next Assy 9344296
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, not applied correctly	1/	5.1	4.8.4
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location		5.1.2	4.6.5/Gauge
Minor				
201	Contents loose		5.1	Manual
202	Workmanship		5.3	Visual
Notes. 1/ Sample size and acceptance criteria shall be in accordance with Table III.				

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4.3.3.15	Title Unitization		Dwg. No. 19-48- 4116/107F	Next Assy —
Classification	Examination Or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical Major	None defined			
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.2	Visual
102	Pallet loading markings – including 2D bar codes- missing incorrect, illegible or not in proper location	VL II	5.2	Visual
103	Filler assembly not built, treated or marked correctly	VL II	5.2	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.2	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.2	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.2	Visual
Minor 201	Workmanship	VL II	5.3	Visual

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4.4.3 Testing.

4.4.3.1 Moisture content. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. If a sample exceeds the requirement, that quantity or sub-lot material represented by the sample-shall not be used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded or packed simulators shall be rejected. Test shall be performed as specified in 4.5.2.

4.4.4 Functioning verifications. The simulator assemblies shall be observed for any evidence of failure to comply with the requirements, as classified in Table III.

TABLE III. Functioning Verification

Defect	Classification
Sound Characteristics (See 3.5.1)	Major
Functioning Time (See 3.5.2)	Major
Signature Characteristics (See 3.5.3)	Major
Cracked Case (See 3.5.4)	Major
No-fire characteristics (See 3.5.5)	Major

4.4.4.1 Hot and ambient temperature. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.4.2.9, fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of 4.4.2.9, thirty-two (32) simulators shall be selected from each lot for this test. The lot shall be rejected if two (2) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.9 and Table IV.

4.4.4.2 Cold temperature. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.4.2.9, fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of - 4.4.2.9, thirty-two (32) simulators shall be selected from each lot for this test. The lot shall be rejected if three (3) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.9 and Table IV.

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TABLE IV. Normal and Reduced Sampling

Temp.	Normal Sampling		Reduced Sampling Plan (After 3 consecutive complying lots)	
	Number of tests	Reject if a critical or any of the following occur	Number of tests	Reject if a critical or any of the following occur
Ambient	50	Three (3) or more major defects	32	Two (2) or more major defects
Hot	50	Three (3) or more major defects	32	Two (2) or more major defects
Cold	50	Four (4) or more major defects	32	Three (3) or more major defects

4.5 Test methods and procedures.

4.5.1 Moisture and volatile content.

4.5.1.1 Signature and bang. The percentage of moisture and volatile content in signature composition shall be determined in accordance with Method 102.2.1 except that the sample shall be heated until a constant weight is obtained in lieu of four hours. Observation shall be made for compliance with the applicable requirement.

4.5.1.1.1 Method 102.2.1. Place the specimen in the tared weighing dish and heat it for 6 hours in the vacuum oven at a temperature of 55° ± 2°C. and a pressure (absolute) of 80 ± 10 mm of mercury. Cool the specimen to room temperature in the desiccator. Determine the loss in weight, and calculate (by weight) the percentage of volatiles in the sample as follows:

Percent volatiles = $\frac{A*100}{W}$

where: A = loss in weight, gm. W = weight of sample, gm.

4.5.2 Functioning. (Ambient Temperature, 21 ± 3°C, 70 ±5°F, Hot, 51.5 ± 3°C, 125°F ± 5°F Cold, -32 ± 3°C -25 ±5°F). The simulator assemblies shall be conditioned for 12 hours min. at the specified temperature and fired within 5 minutes after removal from the conditioning chamber. The simulator shall be mounted in a fixture (simulating the Anti-tank, Missile Fire) on a rigid support and fired by means of a power source in accordance with 3.5 and firing device (drawing 11749562). (Destructive test).

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4.5.2.1 Sound characteristics. Measurement of the sound intensity shall be taken at right angle to the front of the test fixture. The test item and sound detector shall be positioned on a common plane 6 ± .2 feet above ground on a clear and unobstructed area where sound cannot reverberate to give false sound readings. Observation shall be made for compliance with the applicable requirement.

4.5.2.2 Functioning time. The timing equipment shall be in accordance with 4.4.4. Observation shall be made for compliance with the applicable requirement.

4.5.2.3 Signature characteristics. The size and duration of the signature may be determined in accordance with standard practice. These methods can include, but not be limited to, visual or photographic processes using a grid background. Observation shall be made for compliance with the applicable requirement.

4.5.2.4 Crack Case. The case shall be inspected visually after functioning at cold, hot and ambient. Observations shall be made for compliance with the applicable requirement and 4.4.2.9.

4.5.2.5 No-fire characteristics. The M22 simulator shall be classified as a dud after two (2) unsuccessful firing signals are sent.

4.5.3 Jumble (FAT Table II, Critical Characteristic) . The simulator assemblies shall be tested in accordance procedures specified in Standard MIL-STD-331 Test 102, except that the duration of the test shall be limited to a minimum of 20 minutes. The assemblies and equipment shall be observed and inspected for compliance with applicable requirements. Any evidence of functioning, loose powder, or component separation shall be classified as a critical defect. (Destructive test).

4.5.4 Contact closure switch. The resistance of the closure switch shall be measured from edge to edge. The ohmmeter shall be equipped with sharp pointed probes which can penetrate the protective coating on the contact switch. The test shall meet the requirements of 3.3.1. Parts tested shall not be returned to the lot. (Destructive test).

4.5.5 Heat seal of barrier bag. The test shall be performed in accordance with the procedure specified in MIL-STD-2073-1, G.4.3.2. After the test, the seal shall be examined for compliance with the applicable requirements. (Destructive Test).

4.5.6 Vacuum chamber leak test. The test shall be performed in accordance with the procedure specified in MIL-STD-2073-1, G.4.2.2.1.2. Test samples shall be observed for compliance with the applicable requirements. (Non-Destructive Test).

4.5.7 Ammunition lot numbering. Visually verify that the ammunition lot number has been applied to each lot as described in MIL-STD-1168.

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5. PACKAGING

5.1 Packaging and marking.

5.1.1 Packing and marking for M22 ammunition box. Packing and marking shall be in accordance with Drawing 9344297.

5.1.2 Packing and marking for wirebound box. Packing and marking shall be in accordance with Drawing 9344296.

5.2 Palletization

5.2.1 M22 Palletization: Unitize IAW 19-48-4116/107F

5.2.2 Palletized Load Marking: Mark unitized load IAW ACV00561

5.2.3 Wood Products Marking (WPM): WPM IAW ACV00831

5.3 Packaging workmanship. All parts, subassemblies, and assemblies shall be free of defects such as cracks and tears, shall be cleaned by being free of dirt, grease, oil, rust, salt deposits, corrosion, and other foreign matter.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the Simulator, Launching, Anti-Tank Guided Missile and Rocket, M22.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Copies should be submitted to: U.S. Army CCDC AC, ATTN: FCDD-ACE-QMB, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified-on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Distribution of ammunition data cards. Distribution of data cards should include the following, U.S. Army CCDC AC, ATTN: FCDD-ACE-QMB, Picatinny Arsenal, New Jersey 07806-5000.

6.5 Humidity control. Pyrotechnic compositions are hydroscopic materials and will deteriorate rapidly when exposed to moisture. It is highly desirable that all the ingredients of the simulator be carefully safe guarded from exposure to moisture and humid atmosphere. For assurance of the satisfactory loading of the simulator it is recommended that the blending and loading building be air conditioned in a manner that will maintain the relative humidity of the atmosphere below 60%.

6.6 Justification of critical defects.

6.6.1 Evidence of loose powder on exterior of assembly. If there is any presence of charge on exterior of assembly then that may result in accidental ignition or inadvertent functioning of the simulator. This will result in an unintentional function of simulator during handling.

6.6.2 Jumble (FAT Table II, Critical Characteristic) . If a simulator functions during the jumble test then it will be unsafe to handle also if powder leaks then it may cause unintentional functioning of the simulator.

6.7 Definitions.

6.7.1 Bang. Used to express or imitate the sound of a sudden loud noise.
6.7.2 Flash. Burst of bright light or a sudden glint.
6.7.3 Signature. Combination of flash and smoke.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Combat Capabilities Development Command Armaments Center, may also include drawings prepared by, and identified as U.S. Army Combat Capabilities Development Command Armaments Center (DEVCOME AC), Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of DEVCOM AC.

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6.9 Change history.

Specification Revision	ECP Number	ERR Release Date
MIL-S-71146	-	21-June-1993
MIL-DTL-71146A	R17Y2006-1 R18K2023 R18Y3001 R19Q2003 R19Y2008 R22Y2009	

6.10 Subject term (key word) listing.

Closure switch
End plate
Flash composition
Separator disc
Pellet assembly
Training

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army – AR

Preparing activity:
Army – AR
(Project: 1370-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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